

K-Means Clustering

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Agenda

1. Discussion Questions
2. Unsupervised Learning and Clustering
3. Common Distance Metrics
4. Scaling
5. K-Means Clustering
6. Optimal number of clusters
7. Pros & cons

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Questions to discuss

1. What is Unsupervised Learning and Clustering?
2. What is K-Means Clustering and how it works?
3. How to find the optimal number of clusters?
4. How to use t-SNE to visualize clusters?

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Unsupervised Learning

- Unsupervised Learning is a class of Machine Learning techniques to find the patterns in data.
- The data given to unsupervised algorithms are not labeled, which means only the input variables (X) are given with no corresponding output variable (y).
- Involves training of an algorithm using information that is neither classified nor labeled.
- No defined dependent and independent variables.
- Patterns in the data are used to identify/group similar observations

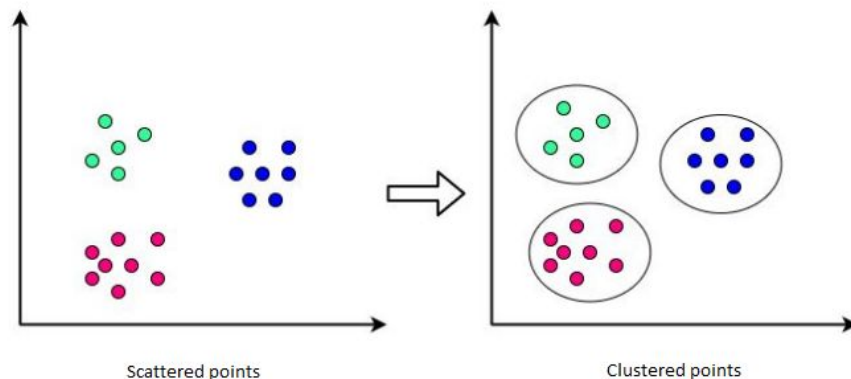
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Clustering

- The objective is to group a set of objects in such a way that objects in the same group are more similar to each other than to those in other groups
- It involves ensuring that the distance between data points in a cluster is very low compared to the distance between 2 clusters.
- This kind of algorithm captures the hidden patterns in data to find the underlying structure and discover new insights.
- The similarity between data points is determined by the distance between them, which can be measured using different distance metrics



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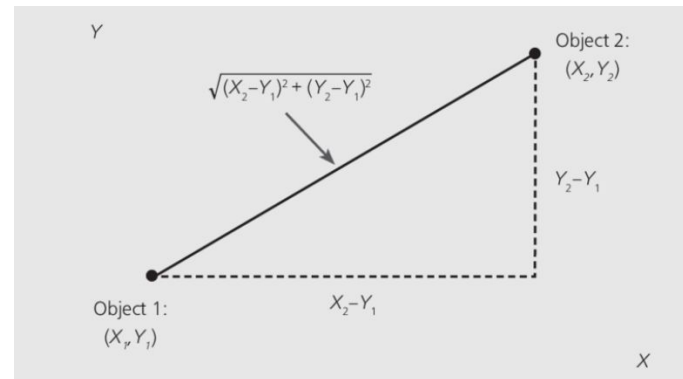
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Common Distance Metrics

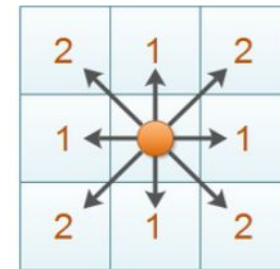
- Euclidean Distance Metrics

$$\sqrt{\sum_{i=1}^k (x_i - y_i)^2}$$



- Manhattan distances

$$\sum_{i=1}^k |x_i - y_i|$$



$$|x_1 - x_2| + |y_1 - y_2|$$

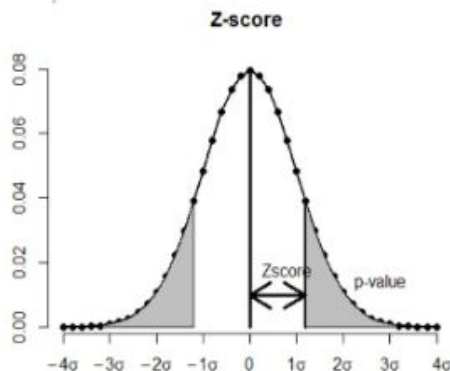
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Scaling the data

- It is important to normalize the data using either Z-score or StandardScaler before performing K-means clustering
- This ensures that the different attributes in the data are of the same scale



$$Z = \frac{x - \mu}{\sigma}$$

Diagram illustrating the Z-score formula with annotations:

- Score**: Points to x
- Mean**: Points to μ
- SD** (Standard Deviation): Points to σ

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K-Means Clustering

- K-Means is one of the most common clustering techniques
- It is a centroid-based clustering algorithm where the objective is to find K clusters / groups
- The working of K-means clustering can be summarized as follows:
 - Step 1: Initialize the K random centroids or K points
 - Step 2: For each data point, calculate the Euclidean distance of it from randomly chosen K centroids and assign each point to a minimum distance cluster.
 - Step 3: Update the centroid by using newly assigned data points to the cluster by calculating the average of data points.
 - Step 4: Repeat the above process for a given no. of iterations or until the centroid allocation no longer changes
- Large K produces smaller groups and small K produces larger groups

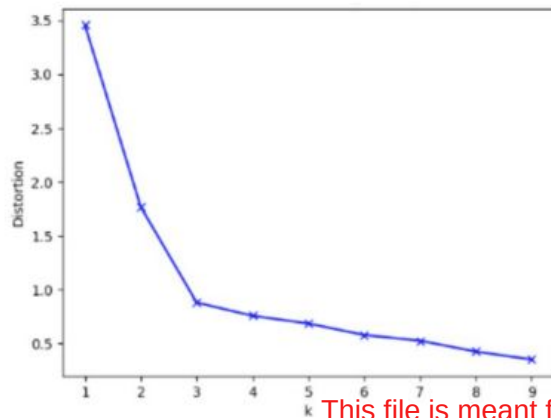
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Optimal Number of Clusters: Elbow Method

- There is no method to define the exact value of K
- The Elbow method is the most popular and well-known method to find the optimal no. of clusters
- This method is based on plotting the value of the cost function against different values of K
- The point where the distortion declines most is said to be the elbow point and defines the optimal number of clusters for the dataset



- In the example here, you can see that the distortion decreases most at 3
- Hence, the optimal value of K will be 3 for performing the clustering

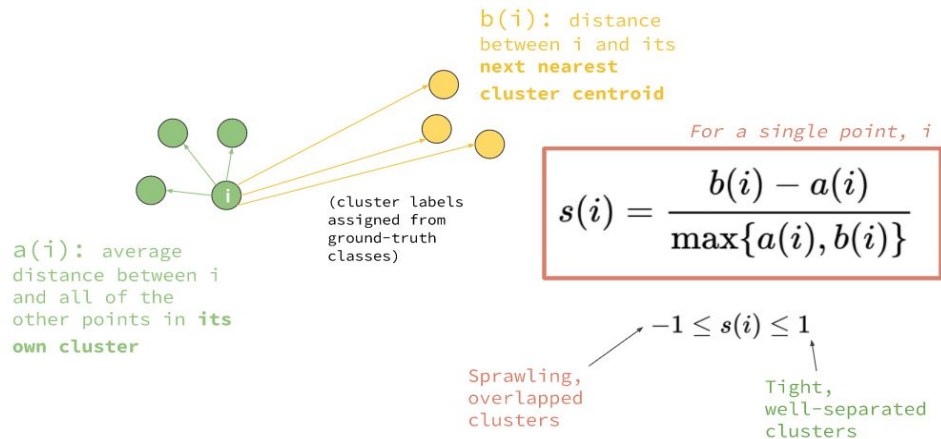
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Optimal Number of Clusters: Silhouette Score

- The silhouette score is a metric which indicates the goodness of clustering algorithms
- It values range between -1 to +1
 - 1 indicates tight, well-separated clusters
 - 0 indicates clusters not well separable
 - -1 indicates data points of one cluster is more closer to centroid of another cluster than the centroid of its own cluster
- **Silhouette score = $(b-a)/\max(a,b)$**
 - a = average intra-cluster distance i.e., the average distance between each point within a cluster.
 - b = average inter-cluster distance i.e., the average distance between all clusters.



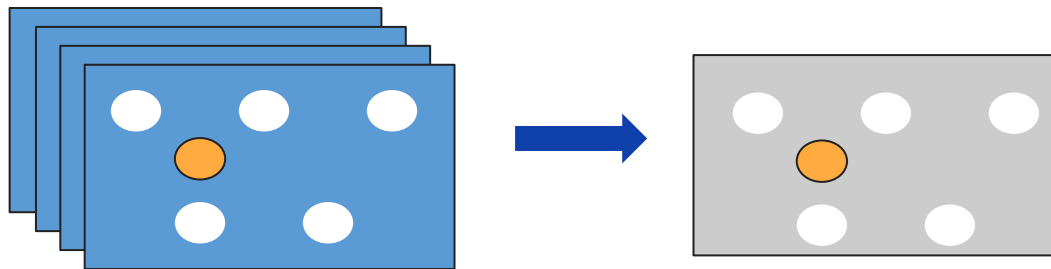
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t-SNE: Overview

- t-SNE stands for **t**-distributed **S**tochastic **N**eighbor **E**mbedding
- Helps visualize high-dimensional data in 2 or 3 dimensions.
- Preserves local data structure when transforming from higher to lower dimensions



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t-SNE: Algorithm

- Initialize with random points in a lower-dimensional space.
- Compute pairwise similarities in the high-dimensional space.
- Compute pairwise similarities in the low-dimensional space.
- Define a loss function that measures the difference between the high-dimensional and low-dimensional similarities.
- Minimize the loss function using optimization techniques (e.g., gradient descent).
- Update the points in the low-dimensional space iteratively.
- Continue until convergence or a set number of iterations.

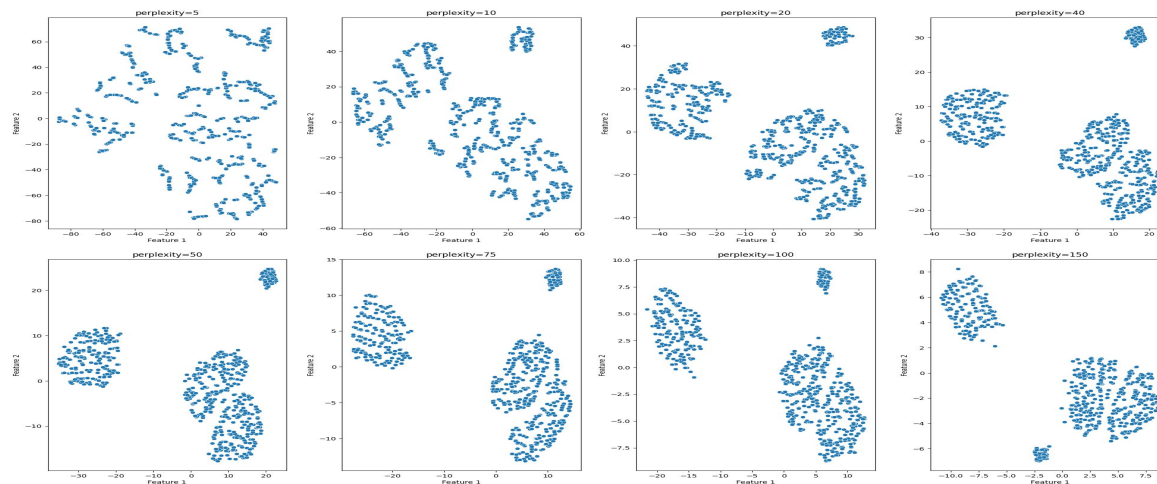
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t-SNE: Perplexity

- A parameter in t-SNE that controls the balance between local and global aspects of data.
- Influences the number of nearest neighbors considered for each data point.
- A low perplexity emphasizes local structure; a high perplexity captures global structure.
- Common range for perplexity values is between 5 and 50.



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